

Dividing Rational Numbers

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Rational numbers are numbers that can be written in the form

$$\frac{a}{b},$$

where a and b are integers and $b \neq 0$.

Reciprocal

To divide rational numbers, multiply the first number by the **reciprocal** of the second number.

The **reciprocal** of a fraction is found by interchanging its numerator and denominator.

$$\frac{3}{4} \div \frac{1}{2} = \frac{3}{4} \times \frac{2}{1}$$

Division Rule

Division of rational numbers follows the rule:

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

where $c \neq 0$ and $d \neq 0$.

Example

Divide $\frac{3}{4}$ by $\frac{1}{2}$.

$$\frac{3}{4} \div \frac{1}{2} = \frac{3}{4} \times \frac{2}{1} = \frac{6}{4} = \frac{3}{2}$$

Mixed Numbers

Convert mixed numbers to **improper fractions** before dividing.

$$2\frac{1}{2} = \frac{5}{2}$$

$$2\frac{1}{2} \div \frac{5}{6} = \frac{5}{2} \times \frac{6}{5} = 3$$

Simplest Form

Always simplify the final answer to its **lowest terms**.

$$\frac{5}{6} \div \frac{10}{9} = \frac{5}{6} \times \frac{9}{10} = \frac{3}{4}$$

Examples

Example 1

Find:

$$\frac{5}{6} \div \frac{10}{9}$$

$$= \frac{5}{6} \times \frac{9}{10} = \frac{3}{4}$$

Example 2

Find:

$$\frac{7}{8} \div \frac{1}{4}$$

$$= \frac{7}{8} \times \frac{4}{1} = \frac{7}{2} = 3\frac{1}{2}$$

Example 3

Find:

$$3\frac{3}{4} \div 4\frac{1}{2}$$

$$= \frac{15}{4} \div \frac{9}{2} = \frac{15}{4} \times \frac{2}{9} = \frac{5}{6}$$

Practice Exercises

Practice 1

Solve. Express your answer in simplest form.

$$\frac{3}{4} \div \frac{1}{2}$$

Practice 2

Solve. Express your answer in simplest form.

$$\frac{5}{6} \div \frac{10}{9}$$

Practice 3

Solve. Express your answer in simplest form.

$$\frac{7}{8} \div \frac{1}{4}$$

Practice 4

Solve. Express your answer in simplest form.

$$\frac{9}{10} \div \frac{3}{5}$$

Practice 5

Solve. Express your answer in simplest form.

$$\frac{2}{3} \div \frac{4}{9}$$

Practice 6

Solve. Express your answer in simplest form.

$$\frac{15}{16} \div \frac{5}{8}$$

Practice 7

Solve. Express your answer in simplest form.

$$\frac{11}{12} \div \frac{22}{15}$$

Practice 8

Solve. Express your answer in simplest form.

$$\frac{7}{9} \div \frac{14}{27}$$

Practice 9

Solve. Express your answer in simplest form.

$$2\frac{1}{2} \div \frac{5}{6}$$

Practice 10

Solve. Express your answer in simplest form.

$$3\frac{3}{4} \div 4\frac{1}{2}$$

Activities

Activity 1

Solve. Express your answer in simplest form.

$$\frac{5}{8} \div \frac{1}{4}$$

Activity 2

Solve. Express your answer in simplest form.

$$\frac{7}{10} \div \frac{14}{15}$$

Activity 3

Solve. Express your answer in simplest form.

$$\frac{9}{10} \div \frac{3}{5}$$

Activity 4

Solve. Express your answer in simplest form.

$$\frac{8}{9} \div \frac{4}{9}$$

Activity 5

Solve. Express your answer in simplest form.

$$\frac{5}{6} \div \frac{10}{9}$$

Activity 6

Solve. Express your answer in simplest form.

$$\frac{21}{20} \div \frac{7}{10}$$

Activity 7

Solve. Express your answer in simplest form.

$$\frac{13}{14} \div \frac{26}{21}$$

Activity 8

Solve. Express your answer in simplest form.

$$\frac{11}{12} \div \frac{22}{36}$$

Activity 9

Solve. Express your answer in simplest form.

$$5\frac{3}{5} \div \frac{4}{5}$$

Activity 10

Solve. Express your answer in simplest form.

$$4\frac{1}{2} \div 2\frac{1}{4}$$

Thank you!